STAYDESK

Phone Call Agent

Product Concept, Call Routing Workflow & MVP Build Guide

Idea-validation stage | Built first for hospitality, designed to expand to businesses globally

Core promise: Human staff answer when available. If they are busy, unavailable, or after-hours, the AI agent takes over the live call so valuable customer calls do not go straight to voicemail.

1. Product Vision

StayDesk Phone Call Agent is a human-first, AI-overflow calling system for businesses. The agent is designed to answer calls when employees are busy, unavailable, after-hours, or handling another customer. The first vertical is hospitality, but the same core calling platform can later support many business categories worldwide.

- Never Miss a Call - automatically take over busy, unanswered, overflow, and after-hours calls.
- Complete the Conversation - answer questions, capture details, create requests, book or qualify when backend rules allow.
- Return Work to the Business - save the outcome, create follow-up tasks, notify staff, and update the dashboard.
- Measure Business Value - track AI-handled calls, recovered booking/lead opportunities, customer outcomes, and verified revenue when attribution is available.

2. Problem We Are Solving

Businesses lose opportunities when customers call and no employee is able to answer. The problem is not only “missed calls.” It is the lost booking, lost lead, delayed service request, poor customer experience, and manual follow-up that can happen after the call is missed.

Current problem	StayDesk approach
Front desk or employee is already on another call	Route the new caller to AI overflow instead of voicemail.
No answer after several rings	Trigger AI after a configurable timeout.
After-hours calls	Send directly to AI according to the business schedule.

Multiple simultaneous calls	Allow the AI layer to handle concurrent overflow calls.
Caller needs a human	Attempt a warm transfer; if nobody is available, continue helping and create follow-up.
Call ends without documentation	Save summary, intent, outcome, transcript/recording metadata (where allowed), and next action.

3. Core Call Flow

#	Stage	What happens
1	**Customer calls existing business number**	The customer uses the same public number the business already advertises.
2	**Human staff gets first chance**	The PBX/carrier rings the front desk or employee for a configured window.
3	**Busy / no-answer / overflow detected**	The phone system applies the routing rule before the call is abandoned.
4	**Call routes to StayDesk**	The live call is forwarded to a programmable number or SIP endpoint connected to the AI agent.
5	**AI understands the customer**	The agent identifies the intent and uses only verified business information and approved backend tools.
6	**AI completes or escalates**	The agent answers, books/creates a request when allowed, or attempts a transfer to staff.
7	**Backend records outcome**	The call session, customer request, action taken, and follow-up status are stored.
8	**Business receives alerts**	The owner/team sees the call in the dashboard and receives alerts for items that need attention.

4. Phone Number Connection Options

Option	How it works	Best for	MVP priority
Existing number + conditional forwarding	Busy/no-answer/after-hours calls forward to the AI destination number.	Fast pilots with small and medium businesses	1 - Start here
New programmable number	Business uses a new number hosted by the telephony provider and controlled by StayDesk.	Testing and businesses willing to use a new number	1 - Also support
SIP/PBX integration	Existing business phone system routes calls directly to StayDesk over SIP.	Larger hotels, franchises, multi-location businesses	2 - After MVP
Number porting / full routing	Business ports or manages its number through the programmable voice stack.	Customers wanting StayDesk to manage the full voice layer	3 - Later

Important: StayDesk should not require special “carrier network access” for the first MVP. The business owner authorizes call forwarding or SIP routing on their existing account. Larger carrier/telecom partnerships can be explored after product validation and customer volume.

5. Technical Architecture

1. Existing phone number / new number
Business telephone provider, or programmable number

2. Routing layer
Busy, no-answer, queue overflow, after-hours, human-first rules

3. Programmable voice gateway
Twilio, Telnyx, Bandwidth, Plivo, or similar provider

4. AI voice agent
Speech/audio session, intent understanding, business conversation

5. StayDesk backend
Authentication, business rules, tools, permissions, routing, logging

6. Business integrations
PMS, CRM, calendar, booking system, task system, payments (read/verified actions only)

7. Database
Calls, customers, leads/bookings, tasks, summaries, outcomes, reporting data

8. Dashboard & alerts
Owner/staff dashboard, email/SMS/app alerts, reports and analytics

6. Agent Backend Tools

The AI agent should not directly edit the database. It should call secure backend tools. The backend validates permissions and business rules before any action is committed.

- get_business_info: Hours, location, services, policies and verified FAQs.
- check_availability: Check rooms, appointments, inventory or service availability from an authorized source.
- create_booking_or_lead: Create a booking request, appointment request, or sales lead after validation.
- create_service_request: Create housekeeping, maintenance, support or other operational tasks.
- lookup_existing_customer: Retrieve permitted customer/reservation context.
- request_human_transfer: Attempt transfer to the right employee or team.
- create_follow_up: Create a staff follow-up when the AI cannot fully complete the request.
- send_confirmation: Send approved email/SMS confirmation through the backend.
- save_call_summary: Save intent, outcome, next action and relevant call metadata.
- send_owner_alert: Notify authorized people about important calls, unresolved issues or opportunities.

7. MVP Features

- Human-first routing with busy/no-answer/after-hours AI takeover.
- Existing number forwarding plus new programmable number support.
- AI agent that answers verified business FAQs and captures customer intent.
- Booking/lead/service-request creation through backend tools.
- Human transfer and fallback workflow.
- Call summary, transcript metadata, outcome, and follow-up status.
- Business dashboard showing all AI-handled calls.
- Owner/staff alerts for urgent or revenue-related calls.
- Basic analytics: total calls, AI overflow calls, inquiries, transfers, follow-ups, abandoned calls.
- Role-based settings for business info, routing rules, hours, escalation numbers and permissions.

8. Business Types We Can Expand Into

Vertical	Typical call workflows	Potential backend action
Hotels & motels	Bookings, policies, guest requests, maintenance	Reservation inquiry, housekeeping task, staff alert
Restaurants	Reservations, hours, large-party questions	Table request, callback, reservation
Dental/medical offices	Scheduling, office info, rescheduling	Appointment request; no clinical diagnosis

Auto repair	Service bookings, vehicle issue intake, status	Service lead, appointment request
Home services	Estimate requests, emergencies, technician scheduling	Lead qualification, service request
Real estate/property management	Leasing inquiries, maintenance, showing requests	Lead, maintenance ticket, appointment
Salons/spas	Appointments, hours, services	Booking request, reschedule
Retail/service businesses	Product/service questions, sales leads	Lead capture, staff follow-up

9. Dashboard UI - Phone Call Agent

AI Overflow Calls **34**	Booking/Lead Inquiries **15**	Human Transfers **4**	Follow-ups Needed **6**
Calls Answered by Staff **46**	Calls Abandoned **0**	Avg AI Call **3m 12s**	Verified Revenue* **$4,280**

*Verified revenue should only be shown when a call can be reliably linked to a confirmed transaction or reservation. Dashboard figures above are illustrative.

10. Testing Before a Real Business Pilot

#	Stage	What happens
1	**Create test phone setup**	Use a programmable test number and a separate human test phone/extension.
2	**Configure human-first routing**	Ring the human first, then forward busy/no-answer calls to the AI.
3	**Run normal calls**	Test business FAQs, booking/lead capture, service requests and follow-ups.
4	**Run failure cases**	Busy line, no answer, after-hours, caller hang-up, AI failure, failed transfer.
5	**Run concurrent calls**	Confirm multiple overflow calls can be handled without dropping sessions.

6	**Verify backend data**	Check every call has the right customer intent, outcome, tasks and alerts.
7	**Review privacy and consent**	Apply applicable call recording, disclosure and data-retention requirements.
8	**Pilot on one business**	Start with limited routing rules and a reliable human/voicemail fallback.

11. Validation Questions for Business Owners

- How many inbound calls do you receive on a normal day?
- How many calls go unanswered, hit voicemail, or are abandoned when staff are busy?
- What percentage of missed calls are sales, bookings, appointments, or customer service?
- What happens today when two or more people call at the same time?
- Do you use busy/no-answer forwarding, a call queue, voicemail, or a call center?
- What information would you trust an AI agent to provide automatically?
- Which actions should always require a human?
- Would you keep your existing number if overflow calls could be routed automatically to AI?
- How would you measure value: fewer missed calls, more bookings/leads, faster service, lower labor cost, or something else?
- Would you consider a paid pilot if we can prove the agent handles overflow calls reliably?

12. First 30-Day Build Plan

Week	Focus	Deliverable
Week 1	Telephony + backend foundation	Programmable number, incoming webhook, call session model, routing configuration.
Week 2	AI conversation + tools	Business knowledge, AI voice session, lead/booking/request tools, summaries.
Week 3	Human overflow + dashboard	Busy/no-answer test flow, human transfer, staff alerts, call dashboard.

Week 4	Reliability + pilot preparation	30-50 controlled test calls, concurrent calls, failure handling, one pilot-ready setup.

13. Product Positioning

“Your staff answers when available. StayDesk takes over when they are not - so important customer calls do not disappear into voicemail.”

Current stage: Idea validation and MVP design. Features, pricing and workflows should be validated with real business owners before production launch.